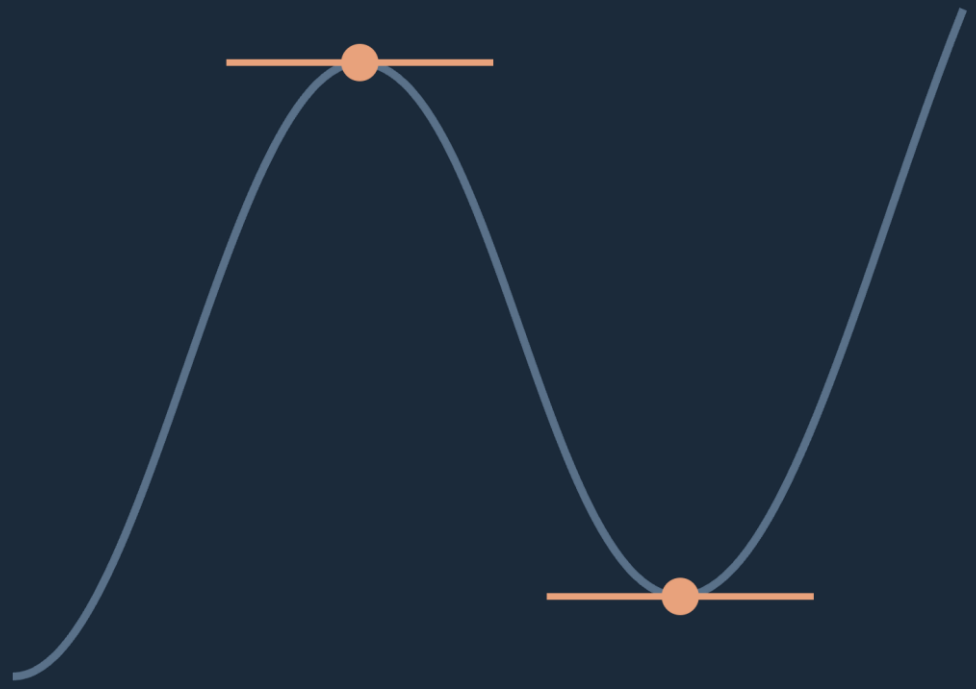
CALCULUS

Applications of Derivatives

Extrema, the Mean Value Theorem and optimization



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A QUESTION TO START

You cut a square from each corner of a 24 × 36 inch sheet and fold up the sides. What size square gives the biggest box?

Cut too little and the box is shallow; cut too much and the base disappears. Somewhere in between is a best answer. Derivatives find it exactly, and we will solve this problem later today.

LEARNING OBJECTIVES

By the end of this lecture you can

01

Tell absolute from local extrema, and find critical points.

02

State and use Rolle's theorem and the Mean Value Theorem.

03

Classify critical points with both derivative tests, and explain why the second one works.

04

Set up and solve optimization problems, and recognize convex functions.

DEFINITION

Absolute extrema

Let f be a function on an interval I, and let c be a point in I.

Absolute maximum at c

$f(c) \geq f(x)$ for every x in I

f(c) is the largest value f takes anywhere on I.

Absolute minimum at c

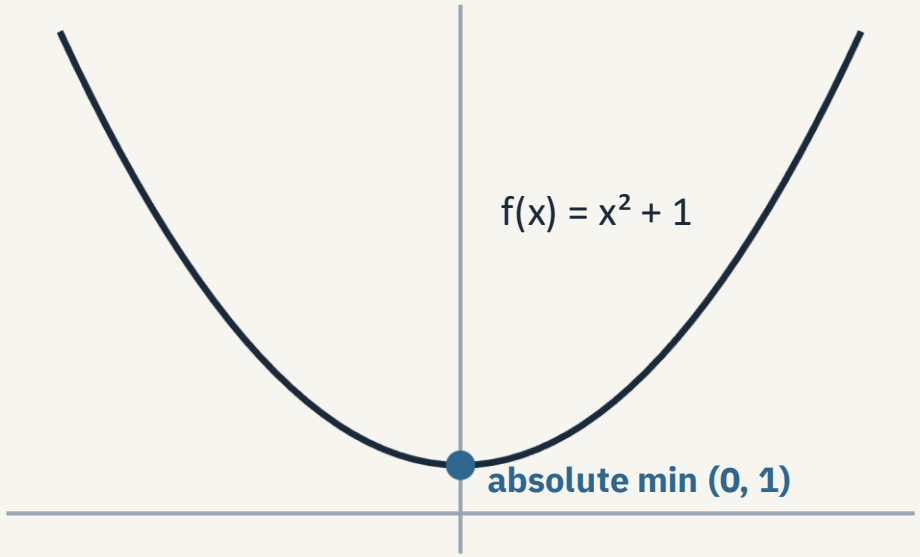
$f(c) \leq f(x)$ for every x in I

f(c) is the smallest value f takes anywhere on I.

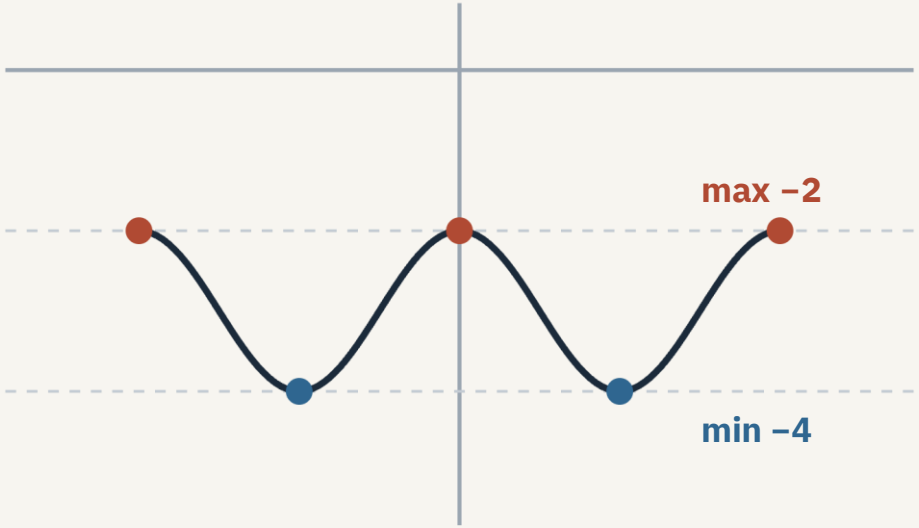
Either one is called an **absolute extremum** of f on I. (Plural: maxima, minima, extrema.)

ABSOLUTE EXTREMA

A function may have one, both or neither



On $(-\infty, \infty)$: an absolute minimum of 1 at $x = 0$, but **no absolute maximum**, because f grows without bound.



$f(x) = -3 + \cos(\pi x/2)$: an absolute maximum of −2 and an absolute minimum of −4, **each reached at several points**.

DEFINITION

Local extrema

"Local" means compared only with nearby points: there is some open interval I containing c, inside the domain of f, where the inequality holds.

Local maximum at c

$f(c) \geq f(x)$ for every x in I

Local minimum at c

$f(c) \leq f(x)$ for every x in I

Either one is a **local extremum**. An absolute extremum at an interior point of the domain is always a local extremum too.

CRITICAL POINTS

Where extrema can hide

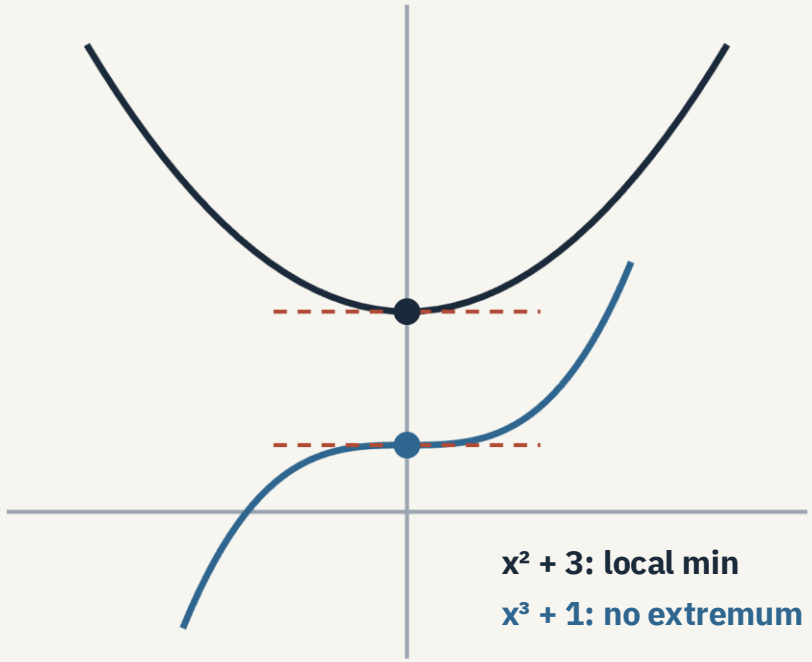
Critical point

An interior point c of the domain of f where **f′(c) = 0** or **f′(c) does not exist**.

Fermat's theorem

If f has a local extremum at c and f is differentiable at c, then f′(c) = 0.

The converse is false: f′(c) = 0 does not guarantee an extremum (blue curve).



KEY CONCEPTS

Four facts about extrema

Existence is not automatic

A function may have both absolute extrema, just one, or none.

Local extrema sit at critical points

If f has a local extremum at c, then c is a critical point.

But not every critical point is one

x^3 has a critical point at 0 and no extremum there.

Extreme Value Theorem

A continuous function on a closed, bounded interval [a, b] has an absolute maximum and an absolute minimum. Each occurs at a critical point or at an endpoint.

EXAMPLE 1

Absolute extrema of $f(x) = x^3 - 3x$ on $[-2, 3]$

1. Find the critical points

$$f'(x) = 3x^2 - 3 = 3(x - 1)(x + 1)$$

Critical points: $x = -1$ and $x = 1$, both inside $[-2, 3]$.

2. Evaluate f at critical points and endpoints

Nothing else can be an absolute extremum.

3. Compare

Largest value: **absolute max 18 at $x = 3$**, an endpoint.
Smallest value: **absolute min -2 at $x = -2$ and $x = 1$**.

x	type	f(x)
−2	endpoint	−2 (min)
−1	critical	2
1	critical	−2 (min)
3	endpoint	18 (max)

THEOREM

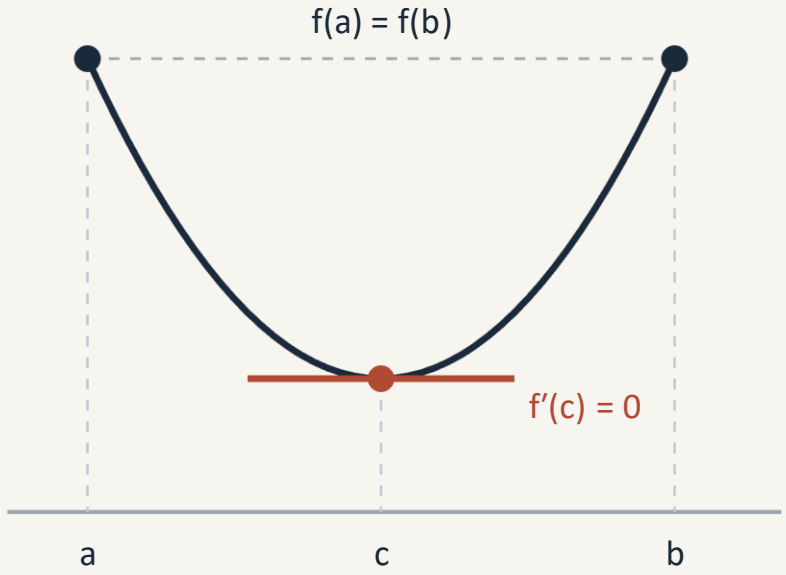
Rolle's theorem

Suppose f is

- continuous on the closed interval [a, b],
- differentiable on the open interval (a, b), and
- f(a) = f(b).

Then there is at least one c in (a, b) with

$$f'(c) = 0.$$



In words: if a smooth curve starts and ends at the same height, it has a horizontal tangent somewhere in between.

THEOREM

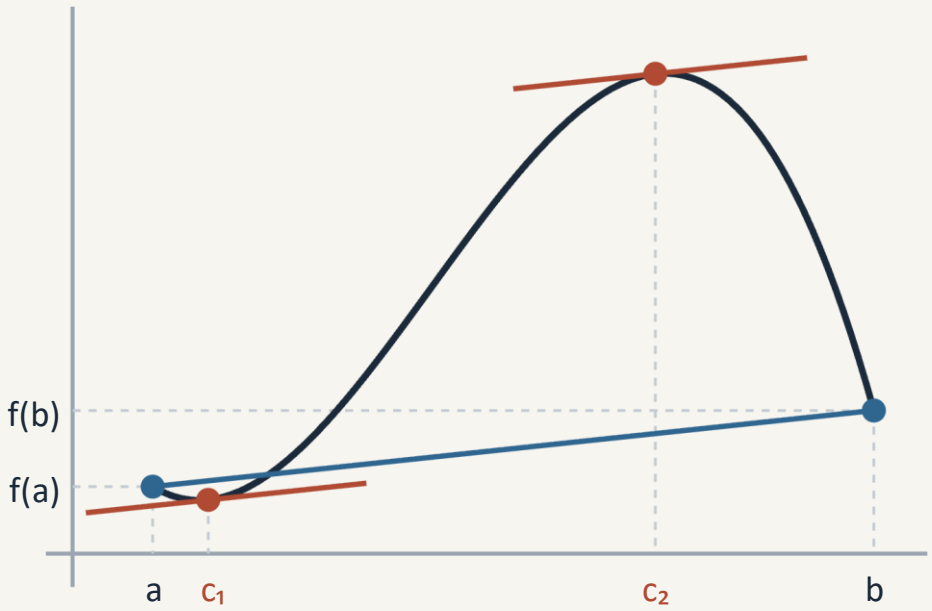
The Mean Value Theorem

If f is continuous on [a, b] and differentiable on (a, b), then there is at least one c in (a, b) with

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Somewhere, the **instantaneous rate** equals the **average rate**: the tangent is parallel to the secant.

Drive 120 miles in 2 hours, and at some moment your speedometer read exactly 60 mph. Rolle's theorem is the special case f(a) = f(b).



MEAN VALUE THEOREM

Three consequences

COROLLARY 1

If $f'(x) = 0$ for every x in an interval I, then f is **constant** on I.

COROLLARY 2

If $f'(x) = g'(x)$ for every x in I, then $f(x) = g(x) + C$ for some constant C.

This is why antiderivatives differ only by a constant.

COROLLARY 3

Let f be continuous on $[a, b]$ and differentiable on (a, b).

$f' > 0$ on $(a, b) \Rightarrow f$ is **increasing** on $[a, b]$.
$f' < 0$ on $(a, b) \Rightarrow f$ is **decreasing** on $[a, b]$.

PART 2

Reading the shape of a graph

The sign of f' tells us where f rises and falls. The sign of f'' tells us which way it bends. Together they classify every critical point.

TEST 1

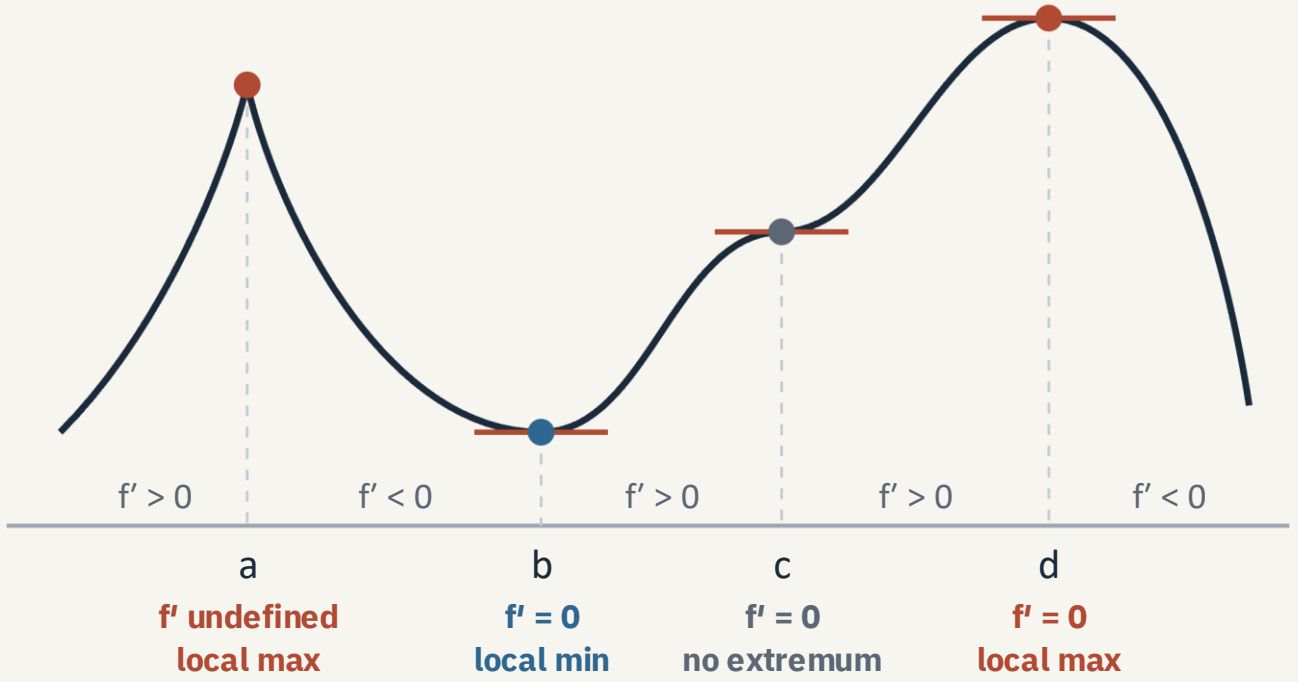
The first derivative test

Let f be continuous on an interval I containing a critical point c, and differentiable on I except possibly at c. Watch the sign of f′ as x passes through c:

+ → −	− → +	no change
rising, then falling	falling, then rising	same sign on both sides
local maximum	**local minimum**	**neither**

TEST 1

Four critical points, three outcomes



METHOD

Finding local extrema in three steps

Let f be continuous on an interval I.

1. **Find the critical points**
   Use them to split I into smaller subintervals.

2. **Find the sign of f′ on each subinterval**
   If **f′** is continuous there, its sign cannot change inside the subinterval, so one test point decides it. A + sign means f is increasing; a − sign means f is decreasing.

3. **Apply the first derivative test**
   At each critical point, decide: local maximum, local minimum, or neither.

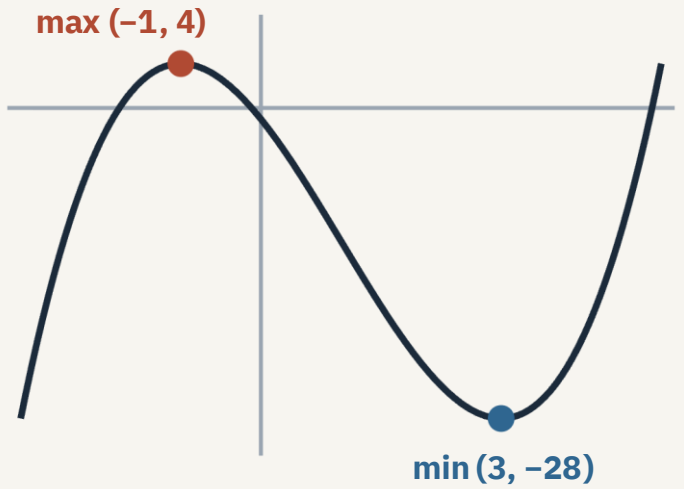
EXAMPLE 2 · FIRST DERIVATIVE TEST

Local extrema of $f(x) = x^3 - 3x^2 - 9x - 1$

$$f'(x) = 3x^2 - 6x - 9 = 3(x + 1)(x - 3)$$

Critical points: $x = -1$ and $x = 3$.

Interval	Test point	Sign of f′	f is
$(-\infty, -1)$	$f'(-2) = 15$	+	increasing
$(-1, 3)$	$f'(0) = -9$	−	decreasing
$(3, \infty)$	$f'(4) = 15$	+	increasing



+ → − at $x = -1$: local max $f(-1) = 4$

− → + at $x = 3$: local min $f(3) = -28$

CONCAVITY

Which way does the graph bend?

Let f be differentiable on an open interval I.



Concave up

Definition: f' is increasing on I.

Test: $f''(x) > 0$ for all x in I.



Concave down

Definition: f' is decreasing on I.

Test: $f''(x) < 0$ for all x in I.

The test needs f to be twice differentiable on I. Picture it this way: a concave-up graph holds water, and it lies above its tangent lines; a concave-down graph spills it, and lies below them.

CONCAVITY

Inflection points

If f is continuous at c and **changes concavity** at c, then (c, f(c)) is an **inflection point**.

Candidates: points where $f''(c) = 0$ or $f''(c)$ does not exist. Then check that f'' really changes sign.

Example: $f(x) = x^3 - 3x^2 - 9x - 1$

$$f''(x) = 6x - 6 = 0 \Rightarrow x = 1$$

$f'' < 0$ for $x < 1$ (concave down) and $f'' > 0$ for $x > 1$ (concave up), so there is an inflection point at **$(1, -12)$**.

Careful: $f''(c) = 0$ is not enough

For $f(x) = x^4$, $f''(0) = 0$, but $f''(x) = 12x^2 \geq 0$ on both sides. The graph never changes concavity, so 0 is not an inflection point.

TEST 2

The second derivative test

Suppose $f'(c) = 0$ and f'' is continuous on an interval containing c.

$f''(c) > 0$	$f''(c) < 0$	$f''(c) = 0$
concave up at a flat point	concave down at a flat point	x^3, x^4 and $-x^4$ all land here at 0
local minimum	**local maximum**	**inconclusive**

When the test is inconclusive, fall back on the first derivative test.

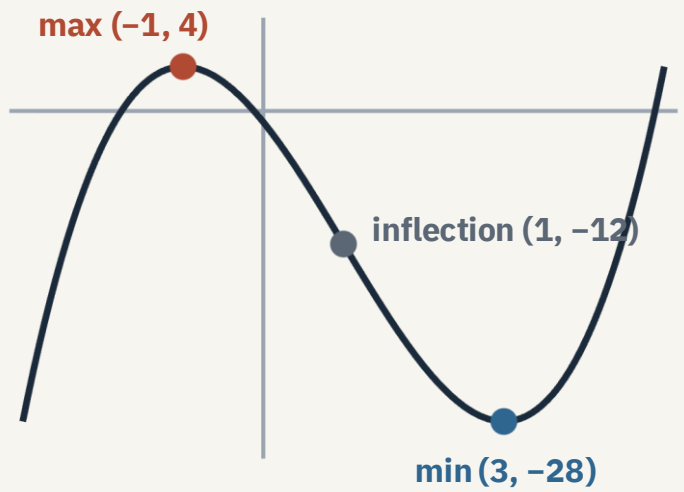
EXAMPLE 3 · SECOND DERIVATIVE TEST

Same function, faster test

$f(x) = x^3 - 3x^2 - 9x - 1$

$f'(x) = 3(x + 1)(x - 3)$, $f''(x) = 6x - 6$

Critical point	f″ there	Conclusion
$x = -1$	$f''(-1) = -12 < 0$	local max, $f(-1) = 4$
$x = 3$	$f''(3) = 12 > 0$	local min, $f(3) = -28$



Same answer as the first derivative test, with one evaluation per critical point instead of a sign table.

QUADRATIC APPROXIMATION

Why the second derivative test works

Add one more term to last lecture's linearization, and near c the curve looks like a parabola:

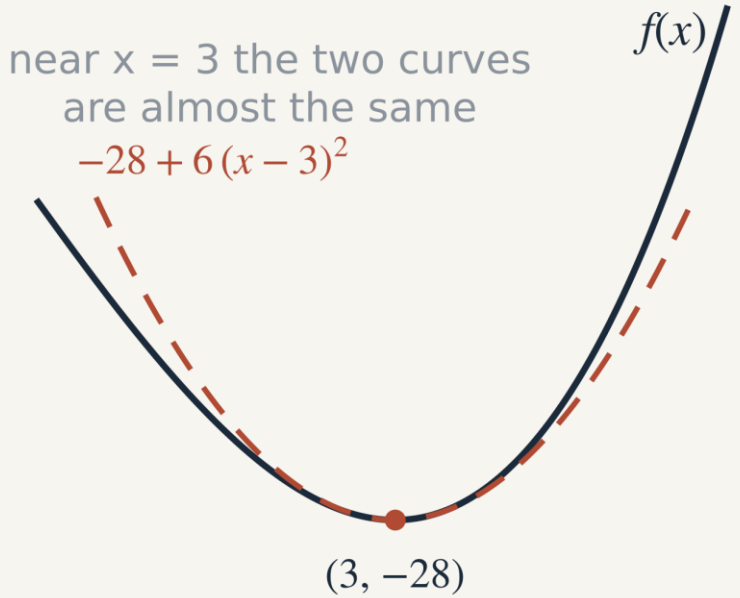
$$f(c + h) \approx f(c) + f'(c)\cdot h + \tfrac{1}{2} f''(c)\cdot h^2$$

At a critical point $f'(c) = 0$, so $f(c + h) \approx f(c) + \tfrac{1}{2} f''(c)\cdot h^2$.

$f''(c) > 0$: the parabola opens up, so c is a local minimum.
$f''(c) < 0$: it opens down, so c is a local maximum.
$f''(c) = 0$: no parabola, so the test says nothing.

Why AI cares

The curvature f'' limits the step size. On $f(x) = x^2$, gradient descent converges only when $\alpha < 2/f'' = 1$.



Example 3 again: $f''(3) = 12$, so near $x = 3$, $f(x) \approx -28 + 6(x - 3)^2$.

SUMMARY

Which test should I use?

Test	What you check	Strengths	Limitations
First derivative	Sign of f' on each side of c	Always decides; works even when $f'(c)$ does not exist	Needs a sign table
Second derivative	Sign of $f''(c)$, when $f'(c) = 0$	One quick evaluation per point	Says nothing when $f''(c) = 0$; needs $f'(c) = 0$

Rule of thumb

If f'' is easy to compute, try the second derivative test first. If it is inconclusive, or f' fails to exist at the point, use the first derivative test.

CONVEXITY

Convex functions: every local minimum is global

A function is convex on an interval if its graph lies above every one of its tangent lines there.

Test: $f''(x) \geq 0$ for every x in the interval

Why it matters: any critical point of a convex function is its global minimum.

Convex: x^2, e^x, $-\ln x$ (for $x > 0$)

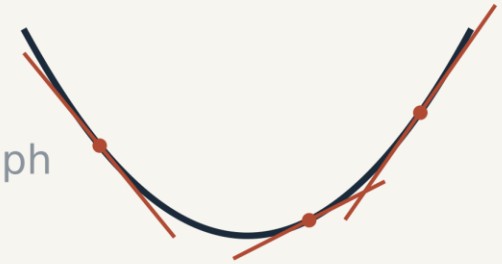
Not convex: $x^3 - 3x$, $\cos x$

Why AI cares

Linear and logistic regression have convex losses, so gradient descent finds the best model.

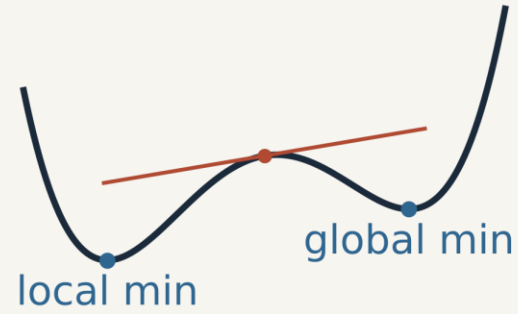
Convex

every tangent line stays below the graph



Not convex

a tangent line crosses the graph



PART 3

Optimization

Optimization means finding the input that makes a quantity as large or as small as possible, usually under some constraints. It appears throughout engineering and economics, and in machine learning, where training a model means minimizing an error function.

METHOD

Solving an optimization problem

Start by introducing variables. If it helps, draw a figure and label everything.

1. **Decide what to maximize or minimize**, and note any limits on the variables.
2. **Write a formula** for that quantity. It may involve several variables.
3. **Use the constraints** to rewrite the formula from step 2 as a function of one variable.
4. **Find the domain** of the function from step 3 that makes physical sense.
5. **Find the max or min** of the function from step 3 over the domain from step 4: check critical points and endpoints.

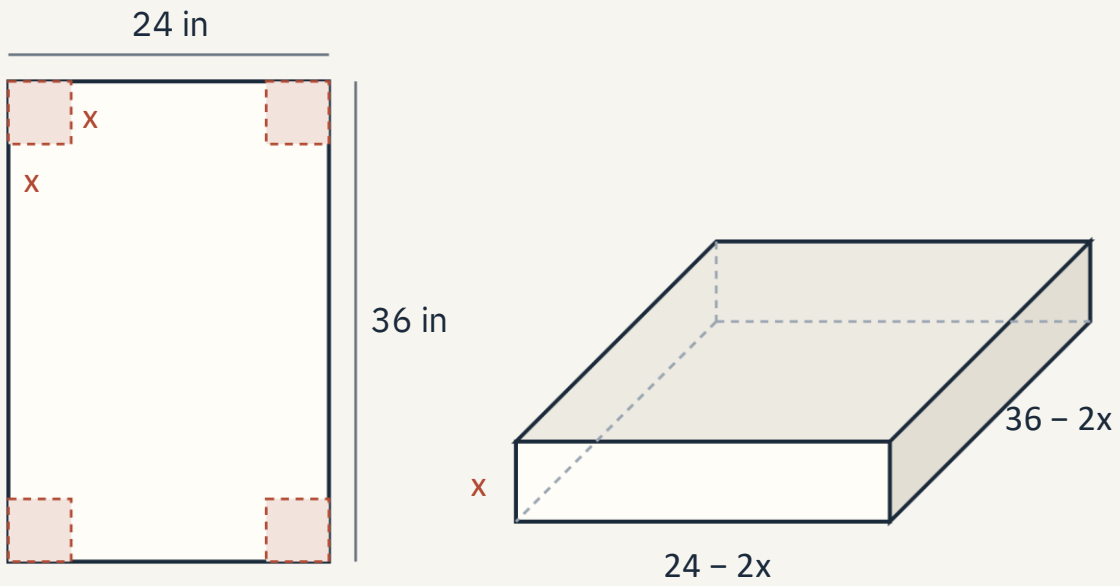
PROBLEM 1

The biggest open box

An open-top box is made from a 24 in × 36 in sheet of cardboard by cutting a square of side x from each corner and folding up the sides.

What size square should be cut out to make the volume as large as possible?

Problem adapted from OpenStax Calculus Volume 1.



PROBLEM 1 · SOLUTION

Cut squares about 4.71 inches wide

$V(x) = x(24 - 2x)(36 - 2x) = 4x^3 - 120x^2 + 864x$ — Steps 1–3: volume

$0 \le x \le 12$, with $V = 0$ at both ends — Step 4: domain

$V'(x) = 12x^2 - 240x + 864 = 12(x^2 - 20x + 72) = 0$ — Step 5: critical points

$x = 10 \pm 2\sqrt{7} \approx 4.71$ or 15.29 — 15.29 is outside the domain

$V''(x) = 24x - 240$, and $V''(4.71) < 0$ — confirms a maximum

$x = 10 - 2\sqrt{7} \approx 4.71$ in, giving a maximum volume of about 1,825 in³

PRACTICE

Your turn

(a) Find and classify the critical points

$f(x) = x^4 - 4x^3$

(b) Find the absolute extrema on [0, 5]

$f(x) = x^3 - 6x^2 + 9x + 1$

(c) Find the c promised by the Mean Value Theorem

$f(x) = x^2$ on $[1, 3]$

(d) Find the concavity and inflection points

$f(x) = x^3 - 6x^2$

(e) Optimization

A farmer has 400 m of fencing to enclose a rectangular field along a straight river. No fence is needed on the river side. What is the largest area she can enclose?

PRACTICE

Answers

(a) $f'(x) = 4x^2(x - 3)$

$x = 0$: $f' < 0$ on both sides, neither. $x = 3$: − to +, local min $f(3) = -27$.

(b) Critical points $x = 1, 3$

$f(0) = 1$, $f(1) = 5$, $f(3) = 1$, $f(5) = 21$. Max 21 at $x = 5$; min 1 at $x = 0$ and $x = 3$.

(c) Average rate $(9 - 1)/(3 - 1) = 4$

$f'(c) = 2c = 4$, so $c = 2$.

(d) $f''(x) = 6x - 12$

Concave down on $(-\infty, 2)$, up on $(2, \infty)$. Inflection point $(2, -16)$.

(e) Sides x, x and $400 - 2x$, with $0 \le x \le 200$

$A(x) = x(400 - 2x)$, $A'(x) = 400 - 4x = 0 \Rightarrow x = 100$. The field is 100 m × 200 m, with area 20,000 m².

WRAP-UP

Key takeaways

Where to look

Local extrema occur only at critical points. On a closed interval, absolute extrema occur at critical points or endpoints.

The big theorem

The Mean Value Theorem: somewhere the instantaneous rate equals the average rate. It explains why the sign of f' tells us where f increases or decreases.

How to classify

First derivative test: a sign change of f'. Second derivative test: the sign of $f''(c)$. It works because $f(c + h) \approx f(c) + \tfrac{1}{2} f''(c)h^2$ near a critical point.

How to optimize

Write one function of one variable, state its domain, then check critical points and endpoints. If f is convex, a critical point is the global minimum.

Thank you. Questions?